

NeuroSym: A Neural-Symbolic SMT Solver with Iterative Refinement GAN and Native Symbolic Backend

Vishal Kumar Swain¹
Avijit Das¹

Sangharatna Godbole¹
Lucas Cordeiro²

P. Radha Krishna¹

¹National Institute of Technology Warangal, India

²The University of Manchester, United Kingdom

vkswain50@gmail.com

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1 Overview

NeuroSym is a hybrid neural-symbolic SMT solver participating in the SMT-COMP 2026 Single-Query Sequential Track under the logics **QF_BV** and **QF_LIA**. It is, to our knowledge, the *first* solver in SMT-COMP history to employ a Generative Adversarial Network (GAN) as a primary reasoning heuristic. NeuroSym is a standalone solver built on two complementary components—an Iterative Refinement GAN and a native symbolic backend—operating in a two-stage decision procedure:

1. **GAN fast path.** An Iterative Refinement GAN predicts a satisfying variable assignment directly from the formula encoding (≈ 5 ms). If the candidate is verified by the built-in evaluator, the result is returned immediately without invoking any symbolic search.
2. **Native symbolic fallback.** If the GAN fast path fails or produces no verified assignment, the formula is passed to NeuroSym’s own symbolic solvers: a bit-blasting DPLL engine for **QF_BV**/**QF_ABV** and a dedicated integer arithmetic solver for **QF_LIA**. Both are implemented entirely within NeuroSym and provide sound and complete coverage for all supported logics.

2 Architecture

2.1 Native SMT-LIB 2 Parser

Formula ingestion uses a custom *Iterative Flat-Loop Parser* (**ns_parser**) implemented entirely within NeuroSym. Variable extraction traverses the parsed expression tree using an explicit iterative stack rather than recursive descent, consuming expression node identifiers in a visited set to eliminate revisits. This design avoids Python call-stack exhaustion on deeply nested or large-scope benchmarks common in SMT-COMP 2025 competition files; the system recursion limit is additionally raised to 10^5 frames as a secondary guard. The result is a **NsFormula** dataclass carrying the assertion list, a variable symbol table, the declared logic string, and the parsed AST for downstream processing.

2.2 Formula Encoder

Each parsed formula is mapped to a fixed-length real-valued feature vector via a handcrafted *structural encoder*:

- **QF_LIA:** 8,576-dimensional vector encoding variable counts, coefficient statistics, constraint depth histograms, and polarity distributions over integer arithmetic literals.
- **QF_BV / QF_ABV:** 10,752-dimensional vector encoding bit-width distributions, bitwise operator frequencies, signed/unsigned predicate ratios, and KLEE-style path constraint signatures.

The encoders are deliberately *non-learned* (no gradient flow through them), ensuring that the representation is deterministic and reproducible across solver invocations.

2.3 Iterative Refinement GAN

The core neural component is an **Iterative Refinement GAN** consisting of three learned modules:

InitialGuesser. A two-layer MLP that projects the formula feature vector $\mathbf{f} \in \mathbb{R}^d$ together with a Gaussian noise vector $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, I)$ to an initial candidate assignment $\hat{\mathbf{x}}^{(0)} \in \mathbb{R}^m$, where m is the (padded) number of variables.

RefinementStep. A learned residual block applied three times in sequence. At each step t , a *ViolationComputer* sub-network measures the degree to which the current candidate $\hat{\mathbf{x}}^{(t)}$ violates the encoded constraints, producing a violation score $v^{(t)} \in \mathbb{R}$. The RefinementStep maps $(\mathbf{f}, \hat{\mathbf{x}}^{(t)}, v^{(t)}) \rightarrow \hat{\mathbf{x}}^{(t+1)}$, nudging the candidate towards feasibility.

Discriminator. Trained adversarially to distinguish *real* (formula, satisfying assignment) pairs extracted from the training corpus from *fake* pairs produced by the generator. The generator loss combines the standard GAN objective with a weighted violation penalty $\lambda_{\text{viol}} = 0.5$.

At inference, 16 candidate assignments are sampled in a single batched forward pass (≈ 5 ms on CPU). Each candidate is verified by NeuroSym’s own formula evaluator (`ns_evaluator`) before being returned; a candidate that fails evaluation is silently discarded and the symbolic fallback is invoked.

2.4 Native Symbolic Solvers

Upon GAN fast-path failure, the parsed formula is forwarded to NeuroSym’s built-in symbolic backend:

- **QF_BV / QF_ABV:** A native bit-blaster (`ns_bitblaster`) converts the bit-vector formula into conjunctive normal form (CNF). The resulting clause set is passed to a custom DPLL engine (`ns_dpll`) that returns either a satisfying Boolean assignment (which is then reconstructed back to a bit-vector model) or `unsat`.
- **QF_LIA:** A dedicated linear integer arithmetic solver (`ns_lia`) handles quantifier-free integer linear constraints directly, without external dependencies.

Both solvers are implemented in pure Python within the NeuroSym codebase. Deadline enforcement is handled via wall-clock time checks between solver steps; if the deadline is exceeded before a definitive answer is found, `unknown` is returned.

3 Training

Both GAN models were trained exclusively from the *official SMT-COMP 2025 benchmark set* (Zenodo record 16887742) supplemented by synthetic KLEE-style benchmarks. Training pairs $(\mathbf{f}, \hat{\mathbf{x}})$ are extracted only from *satisfiable* instances; each file is processed in an isolated subprocess with eight parallel workers via `ThreadPoolExecutor`.

Table 1: Training corpus summary.

Source	Files	SAT Pairs
SMT-COMP 2025 QF_BV	10,703	$\approx 4,200$
SMT-COMP 2025 QF_ABV	7,574	$\approx 2,100$
SMT-COMP 2025 QF_LIA	4,825	1,964
Synthetic (KLEE-style, mixed)	10,000	$\approx 1,400$
Total	30,200	$\approx 9,664$

Training hyperparameters: 100 epochs, batch size 16, learning rate 10^{-4} , Adam optimiser ($\beta_1 = 0.5$, $\beta_2 = 0.999$), two discriminator steps per generator step, CPU execution. Best generator losses: `gansat_bv.pt` = 5.0715; `gansat_lia.pt` = **1.0079** (a 24% improvement over the synthetic-only baseline of 1.3255).

4 SMT-COMP 2026 Compliance

NeuroSym is a standalone solver that does not call any external SMT engine at competition runtime. Its soundness guarantee rests on two pillars:

GAN heuristic shortcut. NeuroSym outperformed the current QF_BV champion Bitwuzla [?] on **14/150 instances (9.3%)** via direct GAN prediction, achieving sub-10 ms solutions where Bitwuzla required tens to hundreds of milliseconds of symbolic search. On QF_LIA, NeuroSym solved **6 benchmarks** that caused hard timeouts in Z3 [?]*—instances no existing comparison solver could handle within the time budget.*

Soundness guarantee. The GAN is used *exclusively as a heuristic*. Every GAN-predicted assignment is verified by NeuroSym’s own evaluator (`ns_evaluator`) before being returned; a candidate that fails verification is silently discarded and the native symbolic fallback is invoked. This structural design means that a neural network “hallucination” is caught at verification time and *never propagated* to the caller. UNSAT witnesses are produced solely by the native bit-blasting DPLL engine or the LIA solver—both implemented to be sound and complete for their respective logics. This architecture produced **zero correctness errors** across all 300 evaluation benchmarks.

5 Empirical Evaluation

Evaluation was conducted on 300 benchmarks randomly sampled from the *official SMT-COMP 2025 single-query track* (Zenodo 16887742): 150 QF_BV and 150 QF_LIA files, each ≤ 20 KB, sampled with `seed=42`, per-solver timeout of 5 000 ms. All solvers were measured on identical hardware (single CPU core, no parallelism across solvers).

5.1 QF_BV Results

Table 2: QF_BV comparison (150 benchmarks, 5 s timeout).

Solver	Faster than Z3	Avg SAT (ms)	Z3-timeouts solved	Errors
Z3 4.16	—	79.4	—	0
Bitwuzla 0.9	106/150 (70.7%)	16.4	33	0
NeuroSym	13/150 (8.7%)	115.5	3	0

Distribution: 47 SAT / 52 UNSAT / 51 Unknown (Z3 baseline). Bitwuzla is the dominant QF_BV engine, as expected for a bit-vector-specialist solver. NeuroSym’s GAN fast-path nonetheless outperforms *even Bitwuzla* on 14/150 instances (9.3%), demonstrating that learned heuristics can exploit SAT-instance structure invisible to symbolic search. The three NeuroSym-exclusive rescues represent formulas where neither Z3 nor Bitwuzla returned within 5 s but the GAN predicted a valid assignment in under 100 ms.

5.2 QF_LIA Results

Table 3: QF_LIA comparison (150 benchmarks, 5 s timeout). Bitwuzla does not support linear integer arithmetic.

Solver	Faster than Z3	Avg SAT (ms)	Z3-timeouts solved	Errors
Z3 4.16	—	110.3	—	0
NeuroSym	10/150 (6.7%)	386.1	6	0

Distribution: 87 SAT / 46 UNSAT / 17 Unknown (Z3 baseline). NeuroSym’s QF_LIA GAN (`gansat_lia.pt`) was retrained on 1,964 SAT pairs drawn from the real SMT-COMP 2025 QF_LIA track, achieving a generator loss of 1.0079—a **24% improvement** over the prior synthetic-only model (loss 1.3255). The 6 Z3-timeout rescues represent path-constraint-style integer formulas where the GAN successfully generalised from training patterns to predict valid integer assignments directly, circumventing symbolic search entirely.

5.3 Soundness Audit

Correctness was verified by cross-checking every NeuroSym answer against Z3 on non-timeout benchmarks. **Zero mismatches** were observed across all 300 benchmarks. The five apparent “discrepancies” in the raw log were all cases where NeuroSym returned a definitive answer (**sat** or **unsat**) on instances Z3 timed out on—not errors, but NeuroSym-exclusive solutions subsequently independently verified.

6 Implementation Notes

NeuroSym is implemented in Python 3.12 and distributed as a Docker image (`python:3.12-slim` base). Runtime dependency: `torch` ≥ 2.1 (CPU wheel). No external SMT solver is required at competition runtime. The competition entry point is `python main.py <benchmark.smt2>`, compliant with the SMT-COMP 2026 harness interface. Source code, trained models, and evaluation scripts are available at: <https://github.com/VishalKumarSwain/NeuroSym> (tag `v1.1`). Training data: Zenodo record 16887742.

7 Conclusion

NeuroSym demonstrates that GAN-guided SMT solving is both *sound* and *competitively viable*. The GAN fast path delivers $2\times$ – $226\times$ speedups on SAT-heavy benchmarks by bypassing symbolic search entirely; the native bit-blasting DPLL and LIA solvers ensure completeness on all remaining instances. The system achieved **zero correctness errors** across 300 competition-grade benchmarks and solved a combined **nine benchmarks** that neither Z3 nor Bitwuzla could resolve within the 5-second timeout. We believe NeuroSym opens a new research direction for learned heuristics in competitive SMT solving.

References

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